DISTRICT COURT
RECRUITMENT
EXAMINATIONS

2026
Edition

DISTRICT COURT
ARITHMETIC

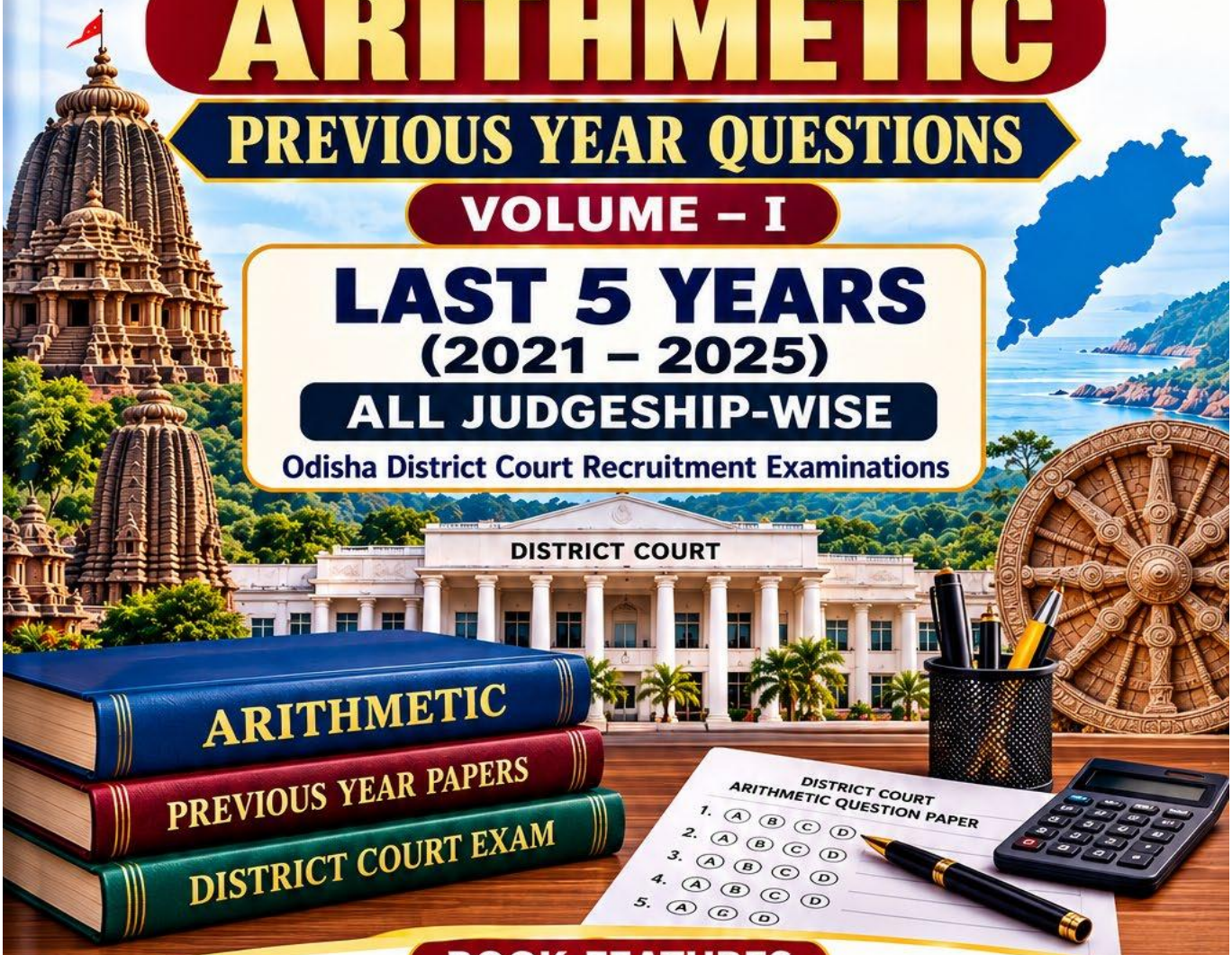
PREVIOUS YEAR QUESTIONS

VOLUME – I

LAST 5 YEARS
(2021 – 2025)

ALL JUDGESHIP-WISE

Odisha District Court Recruitment Examinations



BOOK FEATURES

- Previous Year Question Papers (2021 – 2025)
- All 26 Judgeship-wise Papers
- Question Type-wise Organisation
- Exam Pattern & Marks Distribution
- Authentic & Exam-Oriented Compilation

Specially Designed for Junior Clerk-cum-Copyist under District Court Recruitment Examinations

ମନେରଖନ୍ତୁ—ଗଣିତ ଶିଖିବା ଏବଂ କୋର୍ଟ ପରୀକ୍ଷା ପାଇଁ ପ୍ରସ୍ତୁତ ହେବା ସମ୍ପୂର୍ଣ୍ଣ ଭିନ୍ନ କଥା!

ଅନେକ ପରୀକ୍ଷାର୍ଥୀ ସାଧାରଣ **Math** ବହିରୁ କେବଳ **MCQ** ସମାଧାନ କରନ୍ତି କିନ୍ତୁ ବିଗତ ବର୍ଷଗୁଡ଼ିକର

ପ୍ରକୃତ ପ୍ରଶ୍ନପତ୍ର ନଦେଖି ପରୀକ୍ଷା ହଲ୍‌ରେ ଦ୍ୱନ୍ଦ୍ୱରେ ପଡ଼ିଯାଆନ୍ତି। ଓଡ଼ିଶା ଜିଲ୍ଲା କୋର୍ଟ

ପରୀକ୍ଷାରେ କେବଳ ଉତ୍ତର ନୁହେଁ, ବରଂ ଷ୍ଟେପ୍ ମାର୍କିଂ (**Step-by-Step Descriptive Steps**) ଉପରେ

ହିଁ ଚୟନ ନିର୍ଭର କରେ। ଏହି ବହିରେ ୨୦୨୧ ରୁ ୨୦୨୫ ପର୍ଯ୍ୟନ୍ତ ସମସ୍ତ ୨୬+ ଜିଲ୍ଲା କୋର୍ଟର

ପ୍ରକୃତ ପ୍ରଶ୍ନପତ୍ରକୁ ଏକତ୍ର ସଜାଯାଇଛି, ଯାହାଦ୍ୱାରା ଆପଣ ପରୀକ୍ଷାର ସଠିକ୍ କାଠିନ୍ୟତା ଓ ବାରମ୍ବାର

ଆସୁଥିବା ପ୍ୟାଟର୍ଣ୍ଣକୁ ବୁଝି ନିର୍ଭୁଲ ଭାବେ ସମାଧାନ କରିବାର ସମ୍ପୂର୍ଣ୍ଣ ଆତ୍ମବିଶ୍ୱାସ ହାସଲ କରିପାରିବେ!

BEFORE YOU SOLVE ARITHMETIC, KNOW WHAT YOUR EXAMINATION EXPECTS!

FIVE YEARS OF ACTUAL DISTRICT COURT ARITHMETIC QUESTION PAPERS — 2021 TO 2025

Are you preparing for the Junior Clerk-cum-Copyist, Junior Typist or Stenographer recruitment examination under Odisha District Courts?
Then you should know one important thing:
Learning Mathematics and preparing for an examination are two different things.

You may know how to calculate percentages, solve fractions, find the HCF and LCM or calculate simple interest. But can you recognise the correct method when an unfamiliar question appears in the examination?

Can you solve a question within the available time?
Have you practised the actual arithmetic questions asked in different Odisha District Courts?
These are some of the challenges faced by aspirants.

THE BIGGEST MISTAKE IN ARITHMETIC PREPARATION

Many students begin solving questions directly from general mathematics books without first understanding the actual standard and nature of questions asked in their target examination.

Some practise only easy questions. Others concentrate on a few familiar chapters and leave important topics untouched.

As a result, they may know mathematics but remain unfamiliar with the variety of questions appearing in District Court recruitment examinations.

THIS VOLUME HELPS YOU BRIDGE THAT GAP.

This book brings together actual Arithmetic question papers from Odisha District Court Recruitment Examinations covering five examination years, from 2021 to 2025. The questions are arranged year-wise and judgeship-wise so that you can practise them in an organised manner.

By working through these papers, you can:

- Understand the actual nature and variety of arithmetic questions.
- Become familiar with the different ways in which mathematical concepts are tested.
- Identify important topics and recurring question patterns.
- Discover your strengths and weaknesses.
- Improve calculation accuracy through regular examination-based practice.
- Develop greater confidence in attempting complete arithmetic papers.

WHAT SHOULD YOU DO FIRST?

Before starting any new chapter from a general mathematics book, take some time to examine the previous-year papers.

Try a few questions yourself. Notice which topics you already understand and which ones require additional learning.

This simple exercise will help you approach your preparation with a clearer direction.

DO NOT PREPARE WITHOUT KNOWING THE EXAMINATION. LET THE ACTUAL QUESTIONS GUIDE YOUR PREPARATION.

COURTPREP ODISHA

LEARN FROM ACTUAL PAPERS. UNDERSTAND THE EXAMINATION. PREPARE WITH PURPOSE.

FIVE YEARS OF ARITHMETIC PAPERS. ONE ORGANISED QUESTION BANK. YOUR JUDGESHIP-WISE PREVIOUS-YEAR PRACTICE RESOURCE

This volume is designed to give you access to Arithmetic questions asked in Odisha District Court Recruitment Examinations across different years and judgeships.

WHAT WILL YOU FIND INSIDE?

Examination Year	Question Collection
2025	Judgeship-wise Arithmetic papers
2024	Judgeship-wise Arithmetic papers
2023	Judgeship-wise Arithmetic papers

2022	Judgeship-wise Arithmetic papers
2021	Judgeship-wise Arithmetic papers

EXPLORE THE ACTUAL QUESTION PATTERNS

The papers in this volume expose you to a wide range of arithmetic topics, including:

1. Number System, HCF and LCM
2. Fractions, Decimals and Simplification
3. Percentage
4. Profit and Loss
5. Simple Interest and Compound Interest
6. Ratio and Proportion
7. Partnership
8. Average
9. Time and Work
10. Speed, Time and Distance
11. Trains
12. Mixture and Alligation
13. Mensuration
14. Problems on Ages and other numerical applications

The questions are drawn from the available examination papers; this list indicates the range of topics represented, not a guarantee that every topic appears in every year.

HOW TO GET THE MAXIMUM BENEFIT FROM THIS BOOK

STEP 1 — START WITH THE LATEST PAPERS

Begin with the 2025 question papers. Observe the question formats, marks distribution and the way arithmetic concepts are presented.

STEP 2 — PRACTISE JUDGESHIP-WISE

Attempt papers from different judgeships. Do not restrict your practice to one district.

STEP 3 — SOLVE WITHOUT LOOKING AT THE SOLUTION

Give yourself a proper time limit. Attempt the questions independently and write down your calculations.

STEP 4 — IDENTIFY YOUR DIFFICULTIES

Mark questions that you could not solve, questions that took too much time and questions in which you made calculation errors.

STEP 5 — LEARN THE REQUIRED CONCEPT

If a question is difficult, identify the underlying mathematical concept and study it from the corresponding chapter of Arithmetic Volume II – Complete Preparation (From Zero to Hero).

STEP 6 — REPEAT AND IMPROVE

After strengthening your concepts, return to the question papers and solve the difficult questions again.

TWO VOLUMES. TWO DIFFERENT PURPOSES. ONE CONNECTED PREPARATION STRATEGY.

VOLUME I	VOLUME II
Previous-year question papers	Complete preparation from Zero to Hero
Actual questions from 2021–2025	Concepts and fundamentals
Year-wise and judgeship-wise practice	Formulas, shortcuts and methods
Understand examination requirements	Learn how to solve different question types
Test your existing knowledge	Strengthen your mathematical foundation

First understand what the examination asks. Then learn how to solve every type of question. Finally, return to the actual papers and test your improvement.

A MESSAGE TO EVERY ASPIRANT

Do not be afraid of Mathematics simply because you have struggled with it in the past. Every difficult question has a concept behind it. Every calculation becomes easier with the right method and sufficient practice.

Use this volume to understand the examination. Use the companion preparation volume to strengthen your concepts. Practise consistently and learn from every mistake.

YOUR TARGET IS NOT JUST TO SOLVE MORE QUESTIONS. YOUR TARGET IS TO SOLVE THEM CORRECTLY, CONFIDENTLY AND WITHIN THE EXAMINATION TIME.

COURTPREP ODISHA

PREPARE SMARTER. PRACTISE REGULARLY. MOVE CLOSER TO YOUR GOAL.

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Balangir Judgeship (2025)
Subject: Arithmetic | **Time:** 1 Hour | **Total Marks:** 100
*(Answer any 12 **Question**s including **Question** No. 16)*

Question 1

The simple interest on a certain sum for 2 years is Rs. 120/- and compound interest is Rs. 129/-. Find the rate of interest.

Question 2

Find the compound interest on Rs. 5,000/- in 2 years at 4% per annum, if the interest being compounded half yearly.

Question 3

Seven bells ring at intervals of 2, 3, 4, 6, 8, 9 and 12 minutes, respectively. They started ringing simultaneously at 5:00 in the morning. What will be the next time when they all ring simultaneously?

Question 4

Rama walking at a speed of 20 km/hr reaches his Office 30 minutes late. Next time, he increases his speed by 4 km/hr but still he is late by 10 minutes. Find the distance of the office from his house.

Question 5

In a container, milk and water are present in the ratio 7:5. If 15 liter water is added to this mixture, the ratio of milk and water becomes 7:8. Find the quantity of water in the new mixture.

Question 6

A reduction of 20% in the price of sugar enables a man to purchase 4 kg more for Rs. 280/-. What is the original rate and what is the reduced rate?

Question 7

A and B can do a piece of work in 12 days, B and C can do it in 15 days, where as C and D can do it in 20 days. In how many days will they finish it working together?
[Note: The original Question paper prints "C and D" by clerical erratum; standard mathematical formulation refers to cyclic pair "C and A".]

Question 8

The average weight of A, B & C is 45 kg. If the average weight of A and B be 40 kg and that of B and C be 43 kg, then what would be the weight of B?

Question 9

Sanjeeb buys goods worth of Rs. 6650, and he gets a rebate of 6% on it. After getting the rebate he pays sales tax @ 10%. Find the amount he will have to pay for the goods?

Question 10

100 Apples are bought at the rate of Rs. 350 and sold at the rate of Rs. 48 per dozen. What is the percentage of profit and loss?

Question 11

A and B enter into a partnership for a year. A contributes Rs. 1500/- and B Rs. 2000/-. After four months they admit C into partnership, who contributes Rs. 2250/-. If B withdraws his contribution after 9 months, how would they share a profit of Rs. 900/- at the end of the year?

Question 12

If 12 men and 16 boys can do a piece of work in 5 days and 13 men and 24 boys can do it in 4 days, how long will 7 men and 10 boys take to do it?

Question 13

A train, 100 meters in length, passes a pole in 10 seconds and another train of the same length travelling in opposite direction in 8 seconds. Find the speed of the second train.

Question 14

Selling his Mobile Phone at Rs. 18000/-, Rakesh made a loss of 10%. Calculate the cost at which he would have sold it to make a profit of 20%.

Question 15

Each side of a square floor is of length 5 meter. Calculate the expenditure for fixing tiles on it at the rate of Rs. 65/- per square meter.

Question 16

Answer any 4 (four) **Question**s out of the following:

- (a) The H.C.F of two numbers is 7 and their L.C.M is 210. If one of the numbers is 35, find the other.
- (b) Convert 1/16 into decimal number and 0.375 into fraction of lowest term.
- (c) Find the largest number by which 9, 27, 36 and 45 are divisible.
- (d) By what least number 9720 be multiplied to get a perfect cube?
- (e) Find the result multiplying 17.365 to 7.85.

Balasore Judgeship (2025)
Subject: Arithmetic | **Time:** 1 Hour | **Total Marks:** 100
*(Answer any 10 **Question**s; each **Question** carries 10 marks)*

Question 1

- (i) Find the least possible 5-digit number, which when divided by 10, 12, 16 and 18 leaves remainder 27?
- (ii) Find the HCF of $\frac{3}{4}$, $\frac{15}{16}$, $\frac{10}{5}$?

Question 2

A man reads 3/8 of a book on a day and 4/5 of the remainder on the second day. If the number of pages still unread is 40, then how many pages did the book contain?

OR

5/12 part of what amount will be equal to 3/4 part of Rs. 100/-?

Question 3

A businessman purchased 30 TV sets in Rs. 300000/-. If the average cost of 15 TV sets is Rs. 9000/-, then find out the average cost of remaining TV sets?

Question 4

The population of a town is 176400. It increases annually at the rate of 5% per annum. What will be its population after 2 years? What was it two years ago?

Question 5

A man sells an article at a profit of 40%. If he had bought it at 40% less and sold for Rs. 5/- less, he would have gained 50%. Find the cost price of the article?

Question 6

Sapna borrowed some money on compound interest and returned it in 3 yr in equal annual instalments. If rate of interest is 15% per annum and annual instalment is Rs. 4,86,680/-, then find the sum borrowed?

Question 7

A mixture of a certain quantity of milk with 16 L of water is worth Rs. 0.75/- per litre. If pure milk be worth Rs. 2.25/- per litre, then how much milk is there in the mixture? *(Water is available free of cost)*

Question 8

A starts a business with Rs. 2000/-. B joins him after 3 months with Rs. 4000/-, C invests a sum of Rs. 10000/- in the business for 2 months only. At the end of year, the business gave a profit of Rs. 112000/-. How should the profit be divided amongst A, B and C?

Question 9

A can do a piece of work in 10 days while B can do it in 15 days. They begin together but 5 days before the completion of the work, B leaves off. Find the total number of days for the work to be completed?

Question 10

A person covers a certain distance with a speed of 18 km/h in 8 min. If he wants to cover the same distance in 6 min, what should be his speed?

Question 11

In an examination, 20% of total number of students failed in History, 15% of total number of students failed in Hindi and 5% of total number of students failed in both. Find the percentage of students who passed in both the subjects?

Question 12

A boy walking at a speed of 20 km/h reaches his school 30 min late. Next time he increases his speed by 4 km/h but still he is late by 10 min. Find the distance of the school from his home?

Bargarh Judgeship (2025)
Subject: Arithmetic | **Time:** 1 Hour | **Full Marks:** 100
*(Answer any 10 **Question**s; each **Question** carries 10 marks)*

Question 1

- (a) Find the HCF of 48, 64, and 80?
- (b) Find the LCM of 18, 24, and 30?

Question 2

- (a) Find the HCF of 256 and 160.
- (b) Find the LCM of 32, 48, and 72.

Question 3

By what percent is 640 greater than 500?

Question 4

In the class 70% of the students read Hindi, 40% students are boys and 20% are girls do not read Hindi. Then what percentage of boys will read Hindi?

Question 5

A man buys two items for Rs. 1500 each. He sells one at 20% profit and another at 15% loss. Find total profit or loss?

Question 6

On selling 17 balls at Rs. 720, there is a loss equal to the cost price of 5 balls. The cost price of a ball is?

Question 7

Find the least number by which 4800 should be multiplied to make it a perfect cube?

Question 8

A invests Rs. 5000, B invests Rs. 7000. After 6 months, C invests Rs. 6000. Find profit ratio at the end of year?

Question 9

A and B are partners in a business. A contributes [1/4] of the capital for 15 months and B received 2/3 of the profit. For how long B's money was used?

Question 10

In how many years will Rs. 8000 amount to Rs. 10400 at 5% SI?

Question 11

Find CI on Rs. 10,000 for 2 years at 8% p.a. compounded annually?

Question 12

Evaluate $\sqrt{12.96}$?

Question 13

A train running at 54 km/h takes 20 seconds to pass a pole. Find the length of the train?

Question 14

A car travels 150 km at 50 km/h. Then it returns the same distance at 75 km/h. What is the average speed?

Boudh Judgeship (2025)
Subject: Arithmetic | **Time:** 1 Hour | **Full Mark:** 100
*(Answer all sections indicating section and **Question** number)*
Section A
*(Answer any four **Question**s; each **Question** carries 10 marks)*

- **Q1.** The HCF of three numbers is 18. When divided by 18, the sum of their quotients is 30. If the numbers are in the ratio 2: 3: 5, find the numbers.
- **Q2.** A car goes from A to B at 60 km/h and returns at 40 km/h. The distance is 120 km one-way. Find the average speed.
- **Q3.** Find the compound interest on Rs. 12,000 for 3 years at 8% per annum, compounded annually. Also find the amount.
- **Q4.** A contractor undertakes to complete a work in 20 days. He employs 25 workers. After 10 days, only 1/3 of the work is done. How many additional workers must be employed to finish the work in time?
- **Q5.** A and B invest Rs. 12,000 and Rs. 18,000 respectively in a business. At the end of the year, they make a profit of Rs. 15,000. What is B's share if they share profits in proportion to their investments?

Section B
*(Answer any three **Question**s; each **Question** carries 10 marks)*

- **Q1.** H.C.F. of 24, 36 and 60 is: (A) 4 (B) 6 (C) 12 (D) 18
- **Q2.** If the principal is Rs. 2,000 and the simple interest for 3 years at 5% per annum. Find the simple interest: (A) Rs. 300 (B) Rs. 250 (C) Rs. 350 (D) Rs. 400
- **Q3.** A shopkeeper buys a pen for Rs. 20 and sells it for Rs. 28. Profit % is: (A) 30% (B) 35% (C) 40% (D) 45%
- **Q4.** Average of 8, 12, 16 and 24 is: (A) 12 (B) 14 (C) 15 (D) 16
- **Q5.** A wall 10 sq. metres costs Rs. 500 to paint. Cost per sq. metre: (A) Rs. 40 (B) Rs. 50 (C) Rs. 60 (D) Rs. 70

Section C
*(Answer any three **Question**s; each **Question** carries 10 marks)*

- **Q1.** A bus travels 180 km at 60 km/h. If it had traveled 20 km/h faster, how much time would it have saved?
- **Q2.** A can complete a work in 12 days. B can complete the same work in 8 days. How long will both take together?
- **Q3.** A shopkeeper sells an article at a 10% loss. If he had sold it for Rs. 108 more, he would have earned a 10% profit. What is the cost price of the article?
- **Q4.** Three partners shared profit in the ratio 2:3:5. If the total profit is Rs. 1,20,000, what is the difference between the shares of the largest and smallest shareholder?
- **Q5.** The ratio of present ages of father and son is 5:2. After 10 years, the ratio will become 2:1. What is the present age of the father?

Cuttack Judgeship (2025)
Subject: Mathematics | **Time:** 1 Hour | **Full Marks:** 100
*(Answer any 10 **Question**s; each **Question** carries 10 marks)*
Question 1

Find the largest number of four digits exactly divisible by 12, 15, 18 and 27.
Question 2

From a group of boys and girls, 15 girls leave. There are then left 2 boys for each girl. After this, 45 boys leave. There are then 5 girls for each boy. Find the number of girls in the beginning.
Question 3

Arun and Sajal are friends. Each has some money. If Arun gives Rs. 30 to Sajal, then Sajal will have twice the money left with Arun. But, if Sajal gives Rs. 10 to Arun, then Arun will have thrice as much as is left with Sajal. How much money does each have?
Question 4

The sum of two numbers is 184. If one-third of the one exceeds one-seventh of the other by 8, find the smaller number.
Question 5

Rajeev's age after 15 years will be 5 times his age 5 years back. What is the present age of Rajeev?
Question 6

The present age of a father is 3 years more than three times the age of his son. Three years hence, father's age will be 10 years more than twice the age of the son. Find the present age of the father.
Question 7

Sixty-five percent of a number is 21 less than four-fifth of that number. What is the number?
Question 8

10% of the inhabitants of a village having died of cholera, a panic set in, during which 25% of the remaining inhabitants left the village. The population is then reduced to 4050. Find the number of original inhabitants.
Question 9

In an examination, 35% of total students failed in Hindi, 45% failed in English and 20% in both. Find the percentage of those who passed in both the subjects.
Question 10

A person incurs 5% loss by selling a watch for Rs. 1140. At what price should the watch be sold to earn 5% profit?
Question 11

A grocer purchased 80 kg of sugar at Rs. 13.50 per kg and mixed it with 120 kg sugar at Rs. 16 per kg. At what rate should he sell the mixture to gain 16%?
Question 12

A bag contains 50 p, 25 p and 10 p coins in the ratio 5: 9: 4, amounting to Rs. 206. Find the number of coins of each type.
Question 13

If 20 men can build a wall 56 metres long in 6 days. What length of a similar wall can be built by 35 men in 3 days?

Question 14

Two pipes A and B can fill a tank in 36 hours and 45 hours respectively. If both the pipes are opened simultaneously, how much time will be taken to fill the tank?

Deogarh Judgeship (2025)
Subject: Arithmetic | **Time:** 1 Hour | **Full Marks:** 100
*(Answer any 10 **Question**s; each **Question** carries equal marks)*

Question 1

The HCF and LCM of two numbers is 6 and 5040 respectively. If one of the numbers is 210, then find the other number.

Question 2

Two vessels A and B contain mixtures of milk and water in the ratios 4:1 and 9:11 respectively. They are mixed in the ratio of 3:2. Find the ratio of milk and water in the resulting mixture.

Question 3

A shopkeeper sold two articles for Rs. 9471/- each. On one, he gained 23% and on the other, he lost 23%. What is the overall percentage gain or loss?

Question 4

The area of a larger circle is 196π square cm, and the area of a smaller circle is 49π square cm. Find the radius of both circles and the difference in their radii.

Question 5

A sum of Rs. 15000/- is invested partly at 12% per annum and the remaining at 10% per annum simple interest. If the total interest at the end of 2 years is Rs. 3344/-, how much money was invested at 10% per annum?

Question 6

A train covers a distance of 576 km at a certain speed. If the speed is decreased by 24 km/hr it will take 2 hours more to cover the same distance. Find $33\frac{1}{3}\%$ of original speed.

Question 7

- (i) Simplify the numerical expression: $[36 \div (-9)] \div [(-24) \div 6]$
- (ii) Find the missing number in the formula: $7 - 24 \div 8 \times m + 6 = 1$

Question 8

A and B can finish a work together in 30 days. B and C can finish the same work together in 24 days and A and C can finish the work together in 40 days. If all three work together, how long will it take them to complete the work?

Question 9

In an examination, 54% of the candidates passed in Science and 42% failed in Mathematics. If 32% failed in both subjects, what percentage passed in both subjects?

Question 10

Check whether 1512 is a perfect cube or not. Also, find the smallest number which must be multiplied by it to make the given number a perfect cube number and find the cube root as well.

Question 11

There is 60% increase in an amount in 6 years at simple interest. What will be the compound interest of Rs. 12,000/- after 3 years at the same rate?

Question 12

A, B, C subscribe Rs. 50,000/- for a business. A subscribes Rs. 4000/- more than B and B Rs. 5000/- more than C. Out of a total profit of Rs. 35,000/-, what is the amount received by A?

Question 13

At present Rama is 15 years older than Shikha. 5 years later Shikha will be 10 years old. What is the Present age of Rama (in years)?

Question 14

Solve the followings:

- (i) If $a^3 + b^3 = 5824$ and $a + b = 28$ then $(a - b)^2 + ab$ is equal to
- (ii) If $(x - 1/x) = 6$, then $(x^3 - 1/x^3)$ is equal to

Question 15

After selling 5 pens a shopkeeper purchases 2 pens with the profit obtained after selling 5 pens. If he earns Rs. 20/- on selling a pen then find selling price of a pen.

Dhenkanal Judgeship (2025)
Subject: Arithmetic | **Time:** 1 Hour | **Total Marks:** 100

Question 1

- (i) Find the smallest number by which 2560 must be multiplied so that the product is a perfect cube (5 marks).
- (ii) Find the square root of the following by long division method: a) 9801 b) 390625 (5 marks)

Question 2

- (i) Find the square root of $2\frac{1}{5}$ correct to two decimal places (5 marks).
- (ii) Find the value of following: a) $(125)^{2/3}$ b) $8^{4/3}$ (5 marks)

Question 3

The value of a refrigerator which was purchased two years ago depreciates at 12% per annum. If its present value is Rs. 9,680/-, for how much was it purchased (10 marks)?

Question 4

The outer diameter of a metallic tube is 14 cm and thickness of the tube is 1 cm. Find the weight of 2 meters long tube if the density of the metal is 7 gm/cm^3 (10 marks).

Question 5

In a camp, there is enough provision for 500 students for 30 days. If 100 more students join the camp, for how many days will the provision last now (10 marks)?

Question 6

The perimeter of a rectangle is 100 meters. If the length is decreased by 2 meters and the breadth is increased by 3 meters, the area increases by 44 square meters. Find the length and breadth of the rectangle (10 marks).

Question 7

A train 400 meters long is running at a speed of 72 km/hour. How much time does it take to cross a telegraph post (10 marks)?

Question 8

Rajat bought a washing machine on credit. If the washing machine costs Rs. 9,000/- and the shopkeeper charges interest at the rate of 6% per annum, calculate the amount and the compound interest Rajat will have to pay after three years (10 marks).

Question 9

The present population of a Town is 28,000. If the population increases at the rate of 5% per annum in the first year and 7% per annum in the second year, find the population after two years (10 marks).

Question 10

The denominator of a rational number is greater than its numerator by 6. If the numerator is increased by 5 and the denominator is decreased by 3, the number obtained is 5/4. Find the rational number (10 marks).